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GitHub link: <https://github.com/ilyasaqsa/Python-lab>

LAB: 01

Task 1:

Make 2-2 programs of each datatype.

● NUMERIC TYPES:

❖ Integer:

```
:  
num1 = 4  
num2 = 8  
sum = num1 + num2  
print("The sum is:", sum)
```

The sum is: 12

```
:  
num = 2  
  
if num % 2 == 0:  
    print("The number is even")  
else:  
    print("The number is odd")
```

The number is even

❖ Float:

```
num1 = 3.5
num2 = 8.5
result = num1 - num2

print("The result is:", result)
```

The result is: -5.0

```
num1 = 1.0
num2 = 5.0

result = num1 / num2

print("The answer is:", result)
```

The answer is: 0.2

❖ Complex:

```
num1 = 4 + 4j
num2 = 5 + 4j
result = num1 + num2
print("The sum is:", result)
```

The sum is: (9+8j)

```
num1 = 6 + 3j
num2 = 2 + 2j

result = num1 * num2

print("The product is:", result)
```

The product is: (6+18j)

● SEQUENCE TYPES:

❖ String

```
: first_name = "Aqsa"  
  last_name = "Ilyas"  
  full_name = first_name + " " + last_name  
  print("Full name is:", full_name)
```

Full name is: Aqsa Ilyas

```
: message = "Hello, world!"  
  print(message)
```

Hello, world!

❖ List:

```
fruits = ["orange", "banana", "watermelon"]  
for fruit in fruits:  
    print(fruit)
```

orange
banana
watermelon

```
numbers = [1, 2, 3]  
numbers.append(5)  
print("Updated list:", numbers)
```

Updated list: [1, 2, 3, 5]

❖ Tuple:

```
] : colors = ("pink", "brown", "blue")
    print("Second color is:", colors[1])
```

Second color is: brown

```
] : fruits = ("grapes", "banana", "cherry")

    for fruit in fruits:
        print(fruit)
```

grapes
banana
cherry

❖ Range:

```
11]: for num in range(1, 6):
      print(num)
```

1
2
3
4
5

```
12]: for num in range(2, 11, 2):
      print(num)
```

2
4
6
8
10

● SET TYPES:

❖ Set:

```
fruits = {"apple", "banana", "mango"}  
  
print("Fruits set:", fruits)
```

Fruits set: {'apple', 'banana', 'mango'}

```
numbers = {1, 2, 3}  
numbers.add(4)  
  
print("Updated set:", numbers)
```

Updated set: {1, 2, 3, 4}

❖ Frozen set:

```
5]: num = frozenset([1,2,3,4])  
  
print("num frozenset:", num)
```

num frozenset: frozenset({1, 2, 3, 4})

```
6]: set1 = frozenset([1, 2, 3])  
set2 = frozenset([3, 4, 5])  
common = set1.intersection(set2)  
  
print("Common items:", common)
```

Common items: frozenset({3})

● MAPPING TYPE:

❖ Dictionary dict:

```
student = {  
    "name": "aqsa",  
    "age": 18,  
    "class": "BS IT"  
}  
  
print("Student Info:", student)
```

Student Info: {'name': 'aqsa', 'age': 18, 'class': 'BS IT'}

```
person = {  
    "name": "aqsa",  
    "city": "Karachi"  
}  
  
print("Name is:", person["name"])
```

Name is: aqsa

● BOOLEAN TYPE:

```
]:  
a = 30  
b = 10  
  
result = a > b  
print("Is a greater than b?", result)
```

Is a greater than b? True

```
]:  
x = 10  
y = 10  
  
print("Are x and y equal?", x == y)
```

Are x and y equal? True

Task 2:

Make up to 5 shapes programs using *

```

print("Square Shape:")
print("* * * * *")
print("* * * * *")
print("* * * * *")
print("* * * * *")
print("* * * * *")
print("Right-Angled Triangle:")
print("*")
print("* *")
print("* * *")
print("* * * *")
print("* * * * *")
print("Inverted Triangle:")
print("* * * * *")
print("* * * *")
print("* * *")
print("* *")
print("*")

```

Square Shape:

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

Right-Angled Triangle:

```

*
* *
* * *
* * * *
* * * * *

```

Inverted Triangle:

```

* * * * *
* * * *
* * *
* *
*

```

```

print("Pyramid Shape:")
print("  *")
print(" * *")
print(" * * *")
print(" * * * *")
print("* * * * *")
print("Diamond Shape:")
print("  *")
print(" * *")
print(" * * *")
print(" * *")
print("  *")

```

Pyramid Shape:

```

  *
 * *
 * * *
 * * * *
 * * * * *

```

Diamond Shape:

```

  *
 * *
 * * *
 * *
  *

```

```

]:

```

Task 3:

Make same shapes you have made in task 2, using * multiple by number.

Program:


```
]: print("Square Shape:")
print("* " * 5)
print("* " * 5)
print("* " * 5)
print("* " * 5)
print("* " * 5)
print("Right-Angled Triangle:")
print("* " * 1)
print("* " * 2)
print("* " * 3)
print("* " * 4)
print("* " * 5)
print("Inverted Triangle:")
print("* " * 5)
print("* " * 4)
print("* " * 3)
print("* " * 2)
print("* " * 1)
print("Pyramid Shape:")
print(" " * 4 + "* ")
print(" " * 3 + "* " * 2)
print(" " * 2 + "* " * 3)
print(" " * 1 + "* " * 4)
print(" " * 0 + "* " * 5)
```

```
print("Diamond Shape:")
print(" " * 4 + "* ")
print(" " * 3 + "* " * 2)
print(" " * 2 + "* " * 3)
print(" " * 3 + "* " * 2)
print(" " * 4 + "* ")
```

A 5x5 grid of asterisks, representing a square array of 25 elements.

```

*
* *
* * *
* * * *
* * * * *

```

```

* * * * *
* * * *
* * *
* *
*

```

```

      *
     * *
    * * *
   * * * *
  * * * * *

```

Lab 2:

Task 1:

Print numbers from 1 to 10 using a for loop.

Program:

```
[1]: for num in range(1, 11):  
      print(num)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

Task 2:

Print all even numbers between 1 and 20 using a while loop.

Program:

```
[2]: num = 2
      while num <= 20:
          print(num)
          num += 2
```

```
2
4
6
8
10
12
14
16
18
20
```

Task 3:

Calculate the sum of numbers from 1 to 100 using a loop.

Program:

```
[3]: total = 0
      for num in range(1, 101):
          total += num

      print(f"The sum of numbers from 1 to 100 is: {total}")
```

```
The sum of numbers from 1 to 100 is: 5050
```

Task 4:

Print the multiplication table of 5 using a loop:

Program:

```
[4]: num = 5
print(f"Multiplication Table of {num}:")

for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

Multiplication Table of 5:

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Task 5:

Find the factorial of a given number using a for loop.

Program:

```
[5]: num = int(input("Enter a number to find its factorial: "))
      factorial = 1
      if num < 0:
          print("Factorial does not exist for negative numbers.")
      else:
          for i in range(1, num + 1):
              factorial *= i

          print(f"The factorial of {num} is: {factorial}")
```

```
Enter a number to find its factorial: 4
The factorial of 4 is: 24
```

Task 6:

Create a list of numbers and print only the odd numbers using a loop.

Program:

```
[6]: numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
      for num in numbers:
          if num % 2 != 0:
              print(num)
```

```
1
3
5
7
9
```

Task 7:

Iterate over a list of fruits and print each item.

Program:

```
fruits = ["apple", "banana", "cherry", "date", "elderberry", "fig", "grape"]  
for fruit in fruits:  
    print(fruit)
```

```
apple  
banana  
cherry  
date  
elderberry  
fig  
grape
```

LAB 4:

Task 1:

Write a python program to take 2 numbers as input and perform all arithmetic operations on them.

Program:

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
print("Modulus (Remainder):", num1 % num2)
print("Exponent (Power):", num1 ** num2)
print("Floor Division:", num1 // num2)
```

```
Enter first number: 2
Enter second number: 7
Addition: 9.0
Subtraction: -5.0
Multiplication: 14.0
Division: 0.2857142857142857
Modulus (Remainder): 2.0
```

Task 2:

Create a function that takes two numbers and return their sum, difference, product, and quotient.

Program:


```

.: def calculate(a, b):
    sum_result = a + b
    difference = a - b
    product = a * b
    if b != 0:
        quotient = a / b
    else:
        quotient = "Undefined (division by zero)"

    return sum_result, difference, product, quotient
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
sum_result, difference, product, quotient = calculate(num1, num2)
print("Sum:", sum_result)
print("Difference:", difference)
print("Product:", product)
print("Quotient:", quotient)

```

Enter first number: 6
 Enter second number: 9
 Sum: 15.0
 Difference: -3.0
 Product: 54.0
 Quotient: 0.6666666666666666

Task 3:

Write a python script to find the remainder when one number is divided by another.

Program:

```

: num1 = int(input("Enter the first number: "))
  num2 = int(input("Enter the second number: "))
  remainder = num1 % num2
  print("The remainder is:", remainder)

```

Enter the first number: 4
 Enter the second number: 9
 The remainder is: 4

Task 4:

Write a program to calculate the area of a circle using the formula:
 $\text{area} = \pi * r^2$.

Program:

```
:  
import math  
r = float(input("Enter the radius of the circle: "))  
area = math.pi * r ** 2  
  
# Showing the result  
print("Area of the circle is:", area)
```

```
Enter the radius of the circle: 7  
Area of the circle is: 153.93804002589985
```

Task 05:

Implement a program that takes a number as input and returns its square and cube using exponentiation.

Program:

```
[27]: num = float(input("Enter a number: "))  
square = num ** 2  
cube = num ** 3  
print("Square of the number is:", square)  
print("Cube of the number is:", cube)
```

```
Enter a number: 5  
Square of the number is: 25.0  
Cube of the number is: 125.0
```

Task 6:

Create a simple calculator in python that allows the user to choose an operation(addition, subtraction, etc.)and inputs two numbers.

Program:

```
#!/usr/bin/env python3
print("Select operation:")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")
choice = input("Enter choice (1/2/3/4): ")
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
if choice == '1':
    print("Result:", num1 + num2)
elif choice == '2':
    print("Result:", num1 - num2)
elif choice == '3':
    print("Result:", num1 * num2)
elif choice == '4':
    if num2 != 0:
        print("Result:", num1 / num2)
    else:
        print("Error: Cannot divide by zero.")
else:
    print("Invalid choice.")

else:
    print("Invalid choice.")
```

```
Select operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter choice (1/2/3/4): 3
Enter first number: 8
Enter second number: 6
Result: 48.0
```

Lab 5:

Task 1:

Basic Task: Write a program that checks if a given number is positive, negative, or zero.

Program:

```
number = float(input("Enter a number: "))

if number > 0:
    print("The number is positive.")
elif number < 0:
    print("The number is negative.")
else:
    print("The number is zero.")
```

```
Enter a number: 2
The number is positive.
```

Task 2:

Intermediate Task: Write a program that takes user input and determines whether it's a even or odd.

Program:

```
] : number = int(input("Enter a number: "))

if number % 2 == 0:
    print("The number is even.")
else:
    print("The number is odd.")
```

```
Enter a number: 7
The number is odd.
```

Task 3:

Advanced Task: Create a program that asks user to print:
"Excellent" if marks are above 80
"Good" if marks are between 60 and 80
"Needs Improvement" if marks are below 60

Program:

```
] : marks = float(input("Enter your marks: "))

if marks > 80:
    print("Excellent")
elif 60 <= marks <= 80:
    print("Good")
else:
    print("Needs Improvement")
```

```
Enter your marks: 55
Needs Improvement
```

Lab 6:

Task 1:

Basic Task: Write a for loop to print the first 10 natural numbers.

Program:

```
for i in range(1, 11):  
    print(i)
```

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

Task 2:

Intermediate Task: Write a while loop that prints numbers from 10 down to 1

Program:

```
i]: i = 10
    while i >= 1:
        print(i)
        i -= 1
```

```
10
9
8
7
6
5
4
3
2
1
```

Task 3:

Advanced Task: Create a program that uses a for loop to iterate over a string and count the number of vowels.

Program:

```
: text = input("Enter a string: ")
vowels = "aeiouAEIOU"
count = 0

for char in text:
    if char in vowels:
        count += 1

print("Number of vowels:", count)
```

```
Enter a string: a
Number of vowels: 1
```

Challenge Task:

Write a program that prints the Fibonacci series up to n terms using a while loop.

Program:

```
n = int(input("Enter the number of terms: "))
a, b = 0, 1
count = 0

while count < n:
    print(a)
    a, b = b, a + b
    count += 1
```

Enter the number of terms: 3

0

1

1